5 Two uniformly charged non-conducting spheres A and B, of diameter 10 cm and 20 cm respectively, are situated in a vacuum. Their centres are separated by a distance of 50 cm, as shown in Fig. 5.1. x is the distance from the centre of sphere A to any arbitrary point along the line joining the centres of the two spheres.

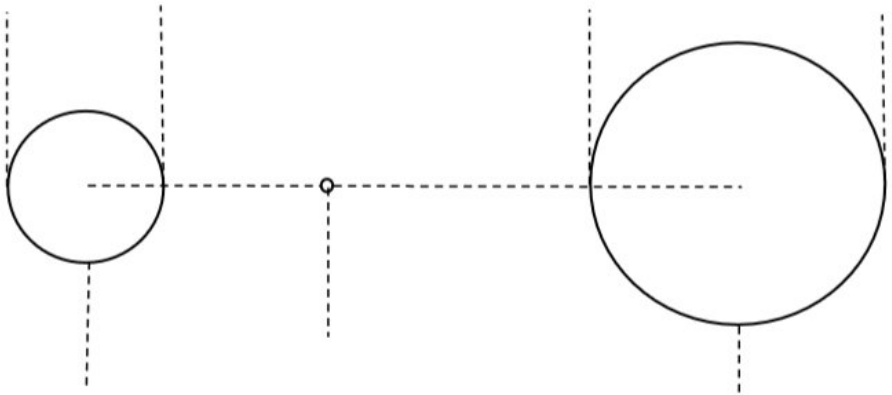


Fig. 5.1

The variation with x of the electric potential V is shown in Fig. 5.2.

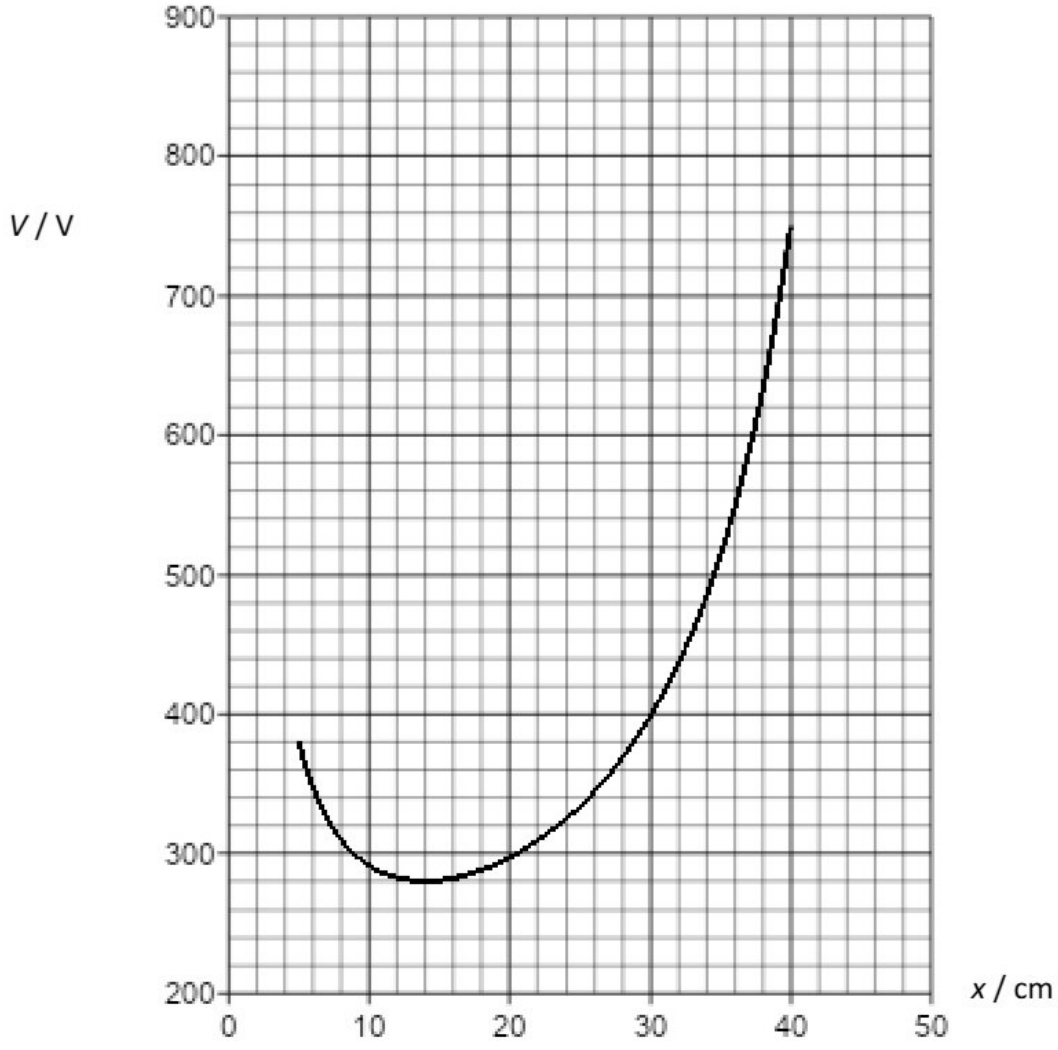


Fig. 5.2

(a) Explain why both charges are positive

.....

..... [1]

(b) Describe how the resultant electric field will vary, along the line joining the centres of the spheres, from the surface of sphere A to the surface of sphere B.

.....

.....

.....

.....

..... [3]

(c) Show the electric field strength at x = 30 cm is 1650 N C^{-1}.

[1]

(d) The charge on each sphere may be considered to be a point charge at its centre. The magnitude of charge of sphere B is 8.0 nC.

Using the information in **(c)**, determine the magnitude of the charge, in nC, of sphere A.

charge of sphere A =..... nC [2]

(e) An electron is ejected from the surface of sphere B. It moves towards sphere A along the line joining the centre of the two spheres.

Determine, the minimum speed that will allow the electron to reach the surface of sphere A.

minimum speed = ... m s ^{1} [3]

[Total: 10]